Week 1: ML Fundamentals & Basic Supervised Models

1. What is Machine Learning (ML)?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers and machines to learn from data, identify patterns, and make predictions or decisions

In simple words, Machine Learning means teaching a computer to learn from previous data and use that learning to make future decisions.

Real-life example

Suppose we want to teach a computer to identify whether an email is spam or not.

1. We provide the computer with thousands of emails.
2. Some emails are labelled as Spam and others as Not Spam.
3. The ML model analyzes patterns, such as suspicious words, links, and sender information.
4. It learns from these examples.
5. When a new email arrives, the model predicts whether it is spam or not.

This is Machine Learning!

Types of Machine Learning

1. Supervised Learning

Definition: Supervised Learning is a type of Machine Learning in which a model learns from labelled data.

Labelled data means the input data already has its correct output or answer.

The model learns the relationship between input and output and uses that knowledge to predict results for new data.

Example: House Price Prediction

Suppose we want to predict house prices based on their size.

House Size (sq. ft.)	Price
1000	$100,000
1500	$150,000
2000	$200,000
2500	$250,000

- Input: House size.
- Output: House price.
- Training: The model learns from existing house sizes and their prices.
- Prediction: If we provide a 3000 sq. ft. house, the model predicts its price.

Main algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

2. Unsupervised Learning

Definition: Unsupervised Learning is a type of Machine Learning in which a model learns from unlabelled data.

Unlabelled data means the data does not contain predefined answers or categories. The model independently identifies hidden patterns, similarities, and relationships.

Example: Customer Segmentation

Suppose a shopping mall has information about 1,000 customers, including:

- Age
- Shopping frequency
- Spending habits

- Purchase history

We provide this data to an ML model without giving it predefined customer groups.

The model analyzes similarities and automatically creates groups:

- Group 1: High-spending customers.
- Group 2: Regular customers.
- Group 3: Low-spending customers.

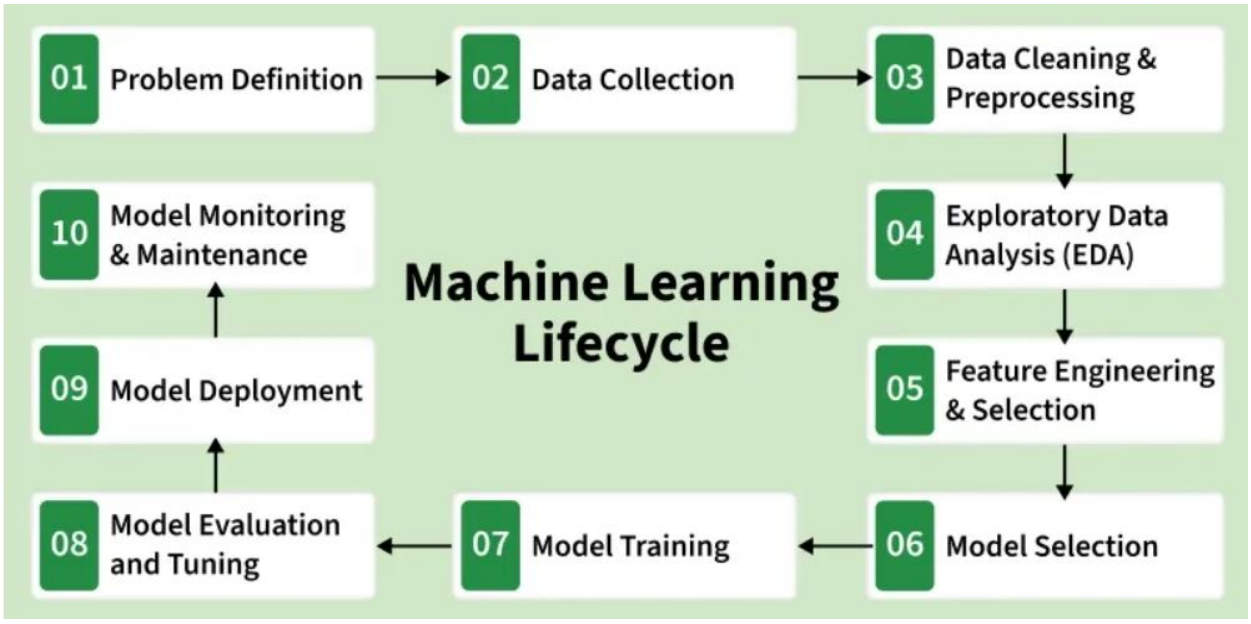
This process is called Clustering.

Main algorithms:

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Apriori Algorithm

2. Machine Learning Lifecycle

Machine Learning Lifecycle is a structured process used to develop, train, deploy and maintain machine learning models efficiently. It includes multiple stages such as data collection, preprocessing, model training, evaluation and monitoring to ensure accurate and reliable predictions.



Step 1: Problem Definition

The first step is clearly defining the problem that needs to be solved. A well-framed problem provides the foundation to determine the project goals, expected outcomes and the type of solution required.

Step 2: Data Collection

Data collection : systematic collection of datasets that can be used as raw data to train model. The quality and variety of data directly affect the model's performance.

Here are some basic features of Data Collection:

- **Relevance:** Collect data should be relevant to the defined problem and include necessary features.
- **Quality:** Ensure data quality by considering factors like accuracy and ethical use.
- **Quantity:** Gather sufficient data volume to train a robust model.
- **Diversity:** Include diverse datasets to capture a broad range of scenarios and patterns.

Step 3: Data Cleaning and Preprocessing

- **Data Cleaning:** Fix missing, incorrect and duplicate data.
- **Data Preprocessing:** Arrange data in a proper format and make values consistent.
- **Data Quality:** Make sure data is clean and ready for analysis.

Step 4: Exploratory Data Analysis (EDA)

- **Exploration:** Study data using graphs and statistics.
- **Patterns and Trends:** Find patterns and trends in the data.
- **Insights:** Understand useful information from the data.
- **Decision Making:** Use these findings to make better decisions.

Step 5: Feature Engineering and Selection

- **Feature Engineering:** Create new useful features from existing data.
- **Feature Selection:** Choose the most important features for the model.
- **Domain Expertise:** Use knowledge of the subject to select useful features.
- **Optimization:** Improve model performance by reducing unnecessary features.

Step 6: Model Selection

- **Complexity:** Choose a model according to the problem and data.
- **Decision Factors:** Consider accuracy, simplicity and performance.
- **Experimentation:** Try different models and select the best one.

Step 7: Model Training

- **Iterative Process:** Train the model again and again to reduce errors.
- **Optimization:** Adjust the model to improve its accuracy.
- **Validation:** Check if the model can work well on new data.

Step 8: Model Evaluation and Tuning

- **Evaluation Metrics:** Check model performance using accuracy, precision, recall and F1 score.
- **Strengths and Weaknesses:** Find out what the model does well and where it needs improvement.
- **Iterative Improvement:** Make changes to improve model performance.
- **Model Robustness:** Make sure the model gives reliable results.

Step 9: Model Deployment

- **Integration:** Connect the model with existing systems.
- **Decision Making:** Use model predictions to make better decisions.
- **Scalability and Security:** Make sure the model is safe and can handle more users or data.
- **APIs and Pipelines:** Make the model available for real-world use.

Step 10: Model Monitoring and Maintenance

- **Performance Tracking:** Regularly check how well the model works.
- **Data Drift:** Detect changes in data that may affect predictions.
- **Retraining:** Train the model again if its accuracy decreases.
- **Logs and Alerts:** Keep records and get alerts when problems occur.

(Source: https://www.geeksforgeeks.org/machine-learning/machine-learning-lifecycle/)

3. The bias–variance tradeoff :

It is a core principle in supervised machine learning and statistics that governs how well a model generalizes to unseen data.

Bias and variance are two sources of error in predictive models. Getting the right balance between the bias and variance tradeoff is fundamental to effective machine learning algorithms. Here is a quick explanation of these concepts:

- **Bias Variance Tradeoff :** Minimizing errors caused by oversimplification and excessive complication requires finding the right balance or tradeoff between the two.
- **Bias:** When a model is too simple and cannot properly learn patterns from data.
- **Variance:** When a model learns the training data too closely and gives poor results on new data.
- **Trade-off:** Reducing bias usually increases variance, and reducing variance usually increases bias.
- **Balance:** We need to balance both to make a model accurate and reliable.

Ways to reduce the problem:

1. Adjust model complexity to balance bias and variance.
2. Increase training data to help the model learn better.
3. Use more data to reduce overfitting and improve predictions on new data.

Algorithm	Bias	Variance
Linear Regression	High Bias	Less Variance
Decision Tree	Low Bias	High Variance
Bagging	Low Bias	High Variance (Less than Decision Tree)
Random Forest	Low Bias	High Variance (Less than Decision Tree and Bagging)

Source: (https://www.bmc.com/resources/topics/bias-variance-machine-learning.html)

4. Exploratory Data Analysis (EDA)

It is an important step in data analysis where we explore, summarize, and visualize data to understand its structure, detect patterns, identify anomalies, test assumptions, and check relationships between variables before applying any machine learning or statistical models.

Types of Exploratory Data Analysis

1. Univariate Analysis

Univariate analysis studies one variable at a time to understand its characteristics and distribution.

Histograms: Show how data values are distributed.

Box plots: Help detect outliers and show data spread.

Bar charts: Used for categorical variables.

2. Bivariate Analysis

Bivariate analysis examines the relationship between two variables to understand how they interact or influence each other. Common techniques include:

Scatter plots: Show the relationship between two numerical variables.

Correlation coefficient: Measures the strength of the relationship between variables.

Cross-tabulation: Displays the relationship between two categorical variables.

Line graphs: Compare two variables over time to identify trends.

Covariance: Shows how two variables change together.

3. Multivariate Analysis

Multivariate analysis studies three or more variables together to understand complex relationships within the dataset. Common techniques include:

Pair plots: Show relationships between multiple variables at once.

Principal Component Analysis (PCA): Reduces dimensionality while preserving important information.

Spatial analysis: Analyzes geographical patterns using maps and location-based data.

Steps for Performing Exploratory Data Analysis

EDA involves a set of steps that help us understand the data, find patterns, detect issues and prepare the data for further analysis or modelling. It can be performed using different tools like:

- **Python:** Pandas for data manipulation, Matplotlib and Seaborn for visualizations and Plotly for interactive charts.



(Source: https://www.geeksforgeeks.org)

5. Gradient Descent (SGD, Batch, Mini-Batch)

Gradient Descent is an optimization technique used to reduce errors in a machine learning model and improve its accuracy.

Think of it like finding the lowest point in a hill by walking downhill.

There are 3 main types:

Batch Gradient Descent: Uses the entire training dataset to update the model.

- Example: If you have 1,000 students' records, it uses all 1,000 records before updating the model.

Stochastic Gradient Descent (SGD): Uses only one data point at a time to update the model.

- Example: It uses one student's record and updates the model, then moves to the next student.

Mini-Batch Gradient Descent: Uses a small group of data points to update the model.

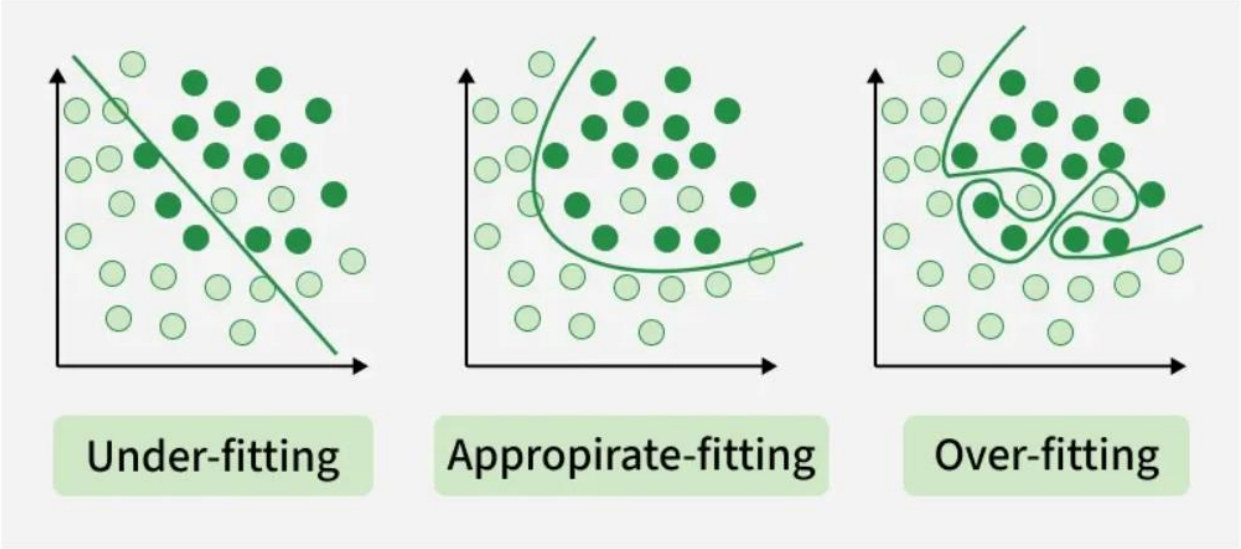
- Example: From 1,000 records, it uses 100 records at a time.

6. Overfitting vs Underfitting

These describe how well a machine learning model learns from training data.

When a model learns too little or too much, we get underfitting or overfitting.

- Underfitting means that the model is too simple and does not cover all real patterns in the data.
- Overfitting means that the model learns not just the underlying pattern, but also noise or random quirks in the training data. model memorizes training data
- A good model finds the right spot, it is complex enough to capture real patterns, but not so complex that it “memorizes” noise



7. Evaluation Metrics

Evaluation metrics are used to check how well a machine learning model performs.

A. Classification Metrics

These are used when predicting categories, such as Pass/Fail or Spam/Not Spam.

1. Accuracy

Measures how many predictions are correct out of all predictions.

Example: Out of 100 emails, the model correctly identifies 90.

Accuracy = 90%.

2. Precision

Measures how many items predicted as positive are actually positive.

Example: The model predicts 20 emails as spam, and 15 are actually spam.

Precision = 15/20 = 75%.

3. Recall

Measures how many actual positive items the model correctly identifies.

Example: There are 20 actual spam emails, and the model finds 15.

Recall = 15/20 = 75%.

4. F1-Score

It is the balance between Precision and Recall.

Example: If Precision is 80% and Recall is 60%, F1-score is approximately 69%.

It is useful when we want both precision and recall to be good.

B. Regression Metrics

These are used when predicting numerical values, such as house prices or temperatures.

5. MSE (Mean Squared Error)

Measures the average squared difference between actual and predicted values.

Example:

- Actual house price = 100,000
- Predicted price = 90,000
- Error = 10,000

MSE squares the error and calculates the average squared error for all predictions.

Lower MSE = Better model.

6. R^2 (R-Squared)

Measures how well a model explains the changes in the target values.

Example: If R^2 = 0.90, it means the model explains 90% of the variation in house prices.

- R^2 = 1: Perfect prediction.
- R^2 = 0: Model explains no variation.
- R^2 closer to 1: Generally better model.